8.4 Configuration Management

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Configuration Management

A software system contains a number of separate components, which have to fit to one another so that the system functions as a whole. The configuration management administers these components.

... it is easier to keep an overview!



better



Chaotic, no structure, orientation is almost impossible

Well-defined structure, orientation is easily possible!

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Configuration Management

- The purpose of configuration management is to establish and maintain the integrity of the component or system, the testware, and their relationships to one another through the project and product lifecycle.
- During test planning, configuration management procedures and infrastructure (tools) should be identified and implemented.



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Configuration Management

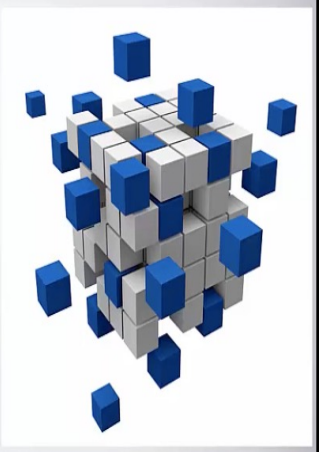
- Objective
  To establish and maintain the integrity (timeliness, completeness) of the products (components, data and documentation) of the software or system through the product and product life cycle
- A software system consists of a number of separate components, which have to fit to one another, so that the system functions as a whole
- At each stage of the system development, every component may have a new, corrected or improved version or variant
- As several developers and testers are involved in this process at the same time, it can be very difficult to maintain an overview of which components are current and belong together; this is part of configuration management

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Configuration Management

- To properly support testing, configuration management may involve ensuring the following:
  - All items of testware are uniquely identified, version controlled, tracked for changes, related to each other and related to versions of the test item(s) so that traceability can be maintained throughout the test process
  - All identified documents and software items are referenced unambiguously in test documentation



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Requirements on Configuration Management (1)

- Version Management
  - Compile, save, recall and retrieve different versions of a configuration object (e.g. version 1.0 and 1.1 of a file)
  - The logging of comments which explain the reason for a change
- Configuration Management
  - Determination and administration of all files (configuration objects) in each version which are built together to make a system (part) configuration
  - A prerequisite for configuration management is version management

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Requirements on Configuration Management (2)

- Following up on the status of incidents and changes
  - Recording of incident reports and change requests
  - Possibility to trace their implementation in the components
- Configuration audits
  - To check the effectiveness of the configuration management, it is reasonable to conduct configuration audits
  - In an audit it can be checked, for example, whether all software components of the configuration management can be gathered and whether their configuration can be identified correctly

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Configuration Management and Testing

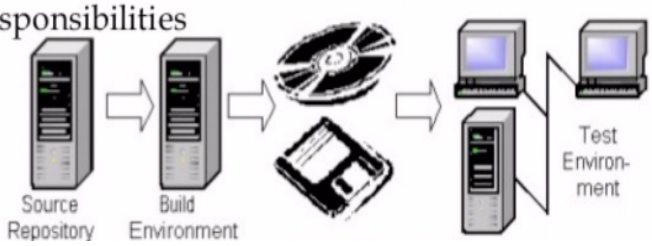
- Configuration management can ensure for testing, that
  - all elements of a testware can be clearly defined and are held under version control
  - changes are traceable and can be related to each other and to the development units (test objects)
  - traceability during the test process or the product life cycle is ensured
  - all documents and software (parts) have a specific allocation within the test documentation
- Configuration management helps the tester to identify (and, if applicable, to reproduce) the test objects, test documents, test cases and the test framework
- During test planning the configuration management procedures and its infrastructure (configuration management tools) are to be selected, described and implemented

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Test Release Management

- Release schedule (i.e., weekly? daily? hourly?)
- Update apply (process to install new build)
- Update unapply (process to remove bad build)
- Build naming (revision level); e.g., X.01.017
- Interrogation (process to determine rev. level)
- Synchronizing with databases, other systems, etc.
- Roles and responsibilities for each step



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JIRA Dashboards RMsis Projects Issues Agile Tests Create Search

RMsis Home | RMsis Administration — Optimizory | admin | Feedback

Requirements | Traceability | Releases | Test Cases | Reporting | Project Configuration — Select Project: Targaryen

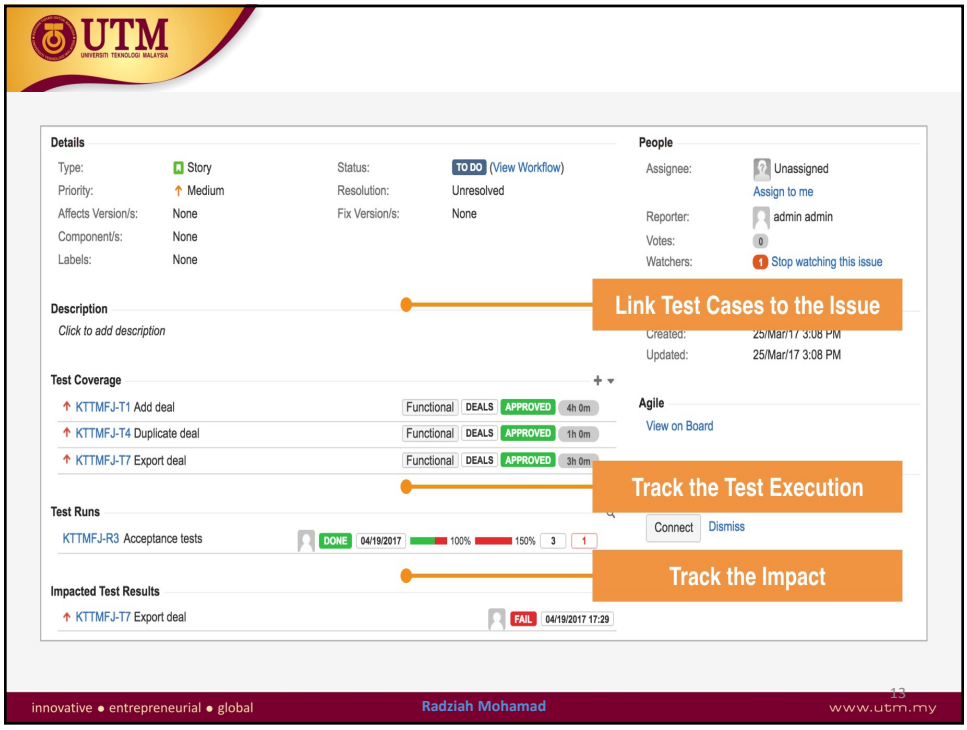
Traceability — Help

Expand All | Collapse All | Latest View | All Releases | Hide Details | CSV Import | Search

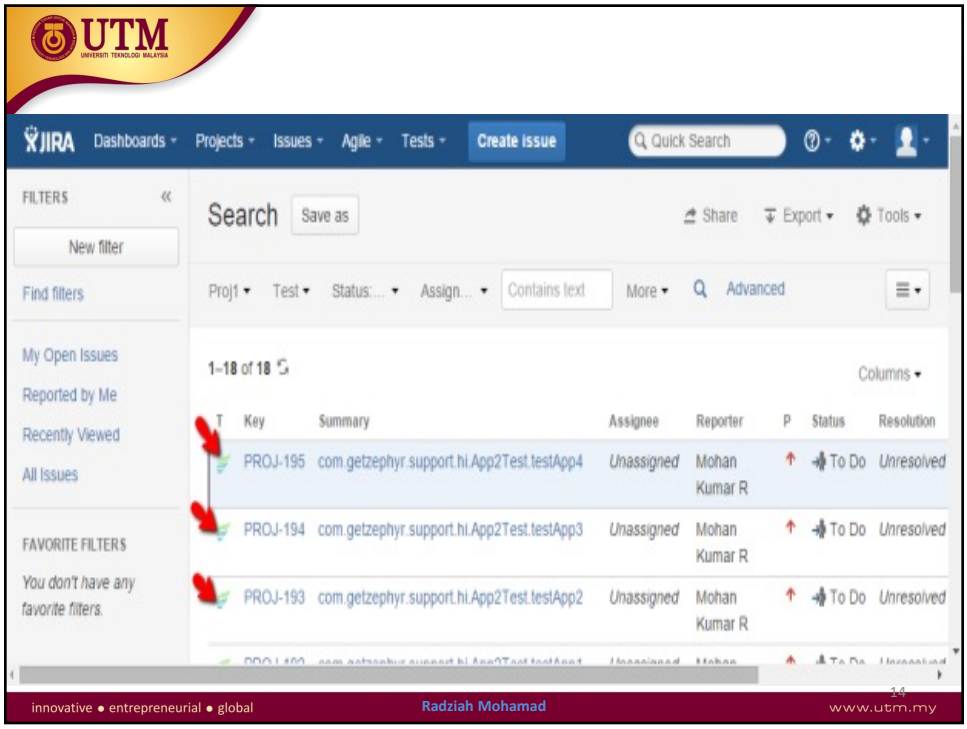
Id	Requirement Summary	Depends On	Dependents	Test Case	Bug	New Feature	Task	Improvement	Sub-task	Requirement
TAR-R16 (1)	Requirement 1	---	---	---	TAR-T1 (1)	---	---	TAR-3	---	---	---
TAR-R1 (1)	Stakeholder Needs	---	DEMO-R1 (1)	DEMO-R2 (2)	---	DEM(35	DEMO-2	DEMO 36	---	---	---
TAR-R4 (1)	User Needs	---	---	---	---	---	---	---	DEMO-3	---	---
TAR-R7 (1)	Product Features	---	---	---	---	---	---	---	---	---	DEMO-22
TAR-R10 (1)	Misc	---	---	---	---	---	---	---	DEMO-3	---	---
TAR-R17 (1)	Requirement 2	---	TAR-R19 (1)	---	---	---	---	---	---	---	DEMO-34
TAR-R18 (1)	Requirement 3	---	TAR-R19 (1)	---	---	---	---	---	---	---	---
TAR-R19 (1)	Requirement 4	---	---	TAR-R17 (1), TAR-R18 (1)	TAR-T2 (1)	---	---	TAR-4	---	---	---
TAR-R20 (1)	Requirement 5	---	---	---	TAR-T3 (1)	---	---	TAR-5	---	---	---
TAR-R21 (1)	Requirement 6	---	---	---	TAR-T4 (1)	---	---	TAR-6	---	---	---
TAR-R22 (1)	Requirement 7	---	---	---	TAR-T5 (1)	---	---	TAR-7	---	---	---
TAR-R23 (1)	Requirement 8	---	---	---	TAR-T6 (1)	---	---	TAR-8	---	---	---

Now showing 20 rows

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Details
Type: Story
Priority: Medium
Affects Version/s: None
Component/s: None
Labels: None
Status: TO DO (View Workflow)
Resolution: Unresolved
Fix Version/s: None
People
Assignee: Unassigned
Assign to me
Reporter: admin admin
Votes: 0
Watchers: 1 Stop watching this issue
Description
Click to add description
Created: 25/Mar/17 3:08 PM
Updated: 25/Mar/17 3:08 PM
Test Coverage
KTTMFJ-T1 Add deal Functional DEALS APPROVED 4h 0m
KTTMFJ-T4 Duplicate deal Functional DEALS APPROVED 1h 0m
KTTMFJ-T7 Export deal Functional DEALS APPROVED 3h 0m
Agile
View on Board
Test Runs
KTTMFJ-R3 Acceptance tests DONE 04/19/2017 100% 150% 3 1
Connect Dismiss
Impacted Test Results
KTTMFJ-T7 Export deal FAIL 04/19/2017 17:29
Link Test Cases to the Issue
Track the Test Execution
Track the Impact

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JIRA Dashboards Projects Issues Agile Tests Create issue Quick Search
FILTERS
New filter
Find filters
My Open Issues
Reported by Me
Recently Viewed
All Issues
FAVORITE FILTERS
You don't have any favorite filters.
Search Save as
Share Export Tools
Proj1 Test Status:... Assign... Contains text More Advanced
1-18 of 18
Columns
T Key Summary Assignee Reporter P Status Resolution
PROJ-195 com.getzephyr.support.hi.App2Test.testApp4 Unassigned Mohan Kumar R To Do Unresolved
PROJ-194 com.getzephyr.support.hi.App2Test.testApp3 Unassigned Mohan Kumar R To Do Unresolved
PROJ-193 com.getzephyr.support.hi.App2Test.testApp2 Unassigned Mohan Kumar R To Do Unresolved

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8.5 Risk and Testing

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Definition of Risk

- Risk involves the possibility of an event in the future which has negative consequences
- The level of risk is determined by the likelihood of the event and the impact (the harm) from that event.



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Risk-Based Testing

The basic idea:

Prioritise tests so that

whenever you stop testing

you have done the best testing

in the time available.

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Product (Quality) Risk

- Product risk involves the possibility that a work product may fail to satisfy the legitimate needs of its users and/or stakeholders, examples include:
  - Software might not perform its intended functions
  - A system architecture may not adequately support some non-functional requirement(s)
  - A particular computation may be performed incorrectly in some circumstances
  - A loop control structure may be coded incorrectly
  - Response-times may be inadequate for a high-performance transaction processing system
  - User experience (UX) feedback might not meet product expectations



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Product Risks

- Potential failure areas in the software or system.
- A risk to the quality of the product.
- Include:
  - Failure prone software delivered.
  - Poor data integrity and quality (e.g. data migration issues, data conversion problems, violation of data standards)
  - Software that does not perform its intended functions.

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Project Risk

- Project risk involves situations that may have a negative effect on a project's ability to achieve its objectives, examples include :
  - Project issues
  - Organizational Issues
  - Political Issues
  - Technical Issues
  - Supplier Issues



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Project Risks

- Organizational factors
  - Skill and staff shortages
  - Personal and training issues
  - Communication problems
  - Lack of follow up on test results, including failure to use such results for process improvement
  - Improper attitude toward or expectations of testing, whether by the testers or by others on the project
  - Complex organization (e.g., distributed)
- Supplier issues
  - Failure of a third party, including a test tool vendor
  - Contractual issues
- Technical issues
  - Problems in defining the right requirements
  - Unachievable requirements (leading to blocked/ unrunnable tests)
  - The quality of the design, code, tests and test data
  - Highly complex system

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Handling Project Risks

- For each project risk, you have 4 options:
  - Mitigation: Reduce the likelihood or impact through preventive steps.
  - Contingency: Have a plan in place to reduce the impact.
  - Transfer: Get some other party to accept the consequences.
  - Ignore: Do nothing about it.

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Risk-Based Testing

- The level of risk varies, depending on:
  - Likelihood arises from technical risk
    - The chance that something might happen.
  - Impact arises from business risk
- In risk-based testing, testing responds to risk:
  - Allocation of effort, test sequencing, prioritization of defect repair.
  - Providing mitigation and contingency responses
  - Reporting test results and project status.

Example of Typical Risk & Its Mitigation and Contingencies

- Logistics or product quality problems block tests
  - ✓ Careful planning, good bug triage, robust test design
- Test deliverables won't install
  - ✓ Smoke testing, nightly builds, uninstall process
- Excessive change invalidates results, requires test updates
  - ✓ Good change control processes, robust test design, limited test documentation, escalation to management
- Insufficient or unrealistic test environment(s)
  - ✓ Explain risks to management, outsource performance testing
- Test environment support unreliable
  - ✓ Good escalation process, sys. admin. skill in team
- Gaps in test coverage revealed during test execution
  - ✓ Exploratory testing, continuous test improvement
- Slips in test start dates and/or deliverables to test
  - ✓ Risk priority to drop tests, escalation to management
- Budget and/or staffing cuts
  - ✓ Risk priority to drop tests, well-rounded team, outsource testing
- Debugging in the test environment
  - ✓ Escalation to management, risk priority to drop/reschedule tests

List of Risk

Risk analysis					Risk status		
ID	**Description**	**Likelihood**	**Impact**	**Total Risk**	**Contingency**	**Responsible**	**Date**
R7	Crash of configuration management database	9	500	4500	Purchase a new server	ABC	15.06.2011
R2	Development delays to software deliveries	9	100	900	Add contingency to test plan	EFG	21.06.2011
R4	Software delivered with	9	10	90	Conduct more code reviews	PQR	25.06.2011

Legend:
Likelihood (1 = negligable, ... , 9 = very high)
Impact (non-linear scale of (money) values (e.g. 1, 10, 50, 100, 500, 1000)
Total Risk = L * I

Table based on:
Spillner, Roßner, Winter, Linz:
Software Testing Practice – Test Management.
Rocky Nook, 2007

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